

Rational - Canonical Form

Jordan - Canonical Form

Smith - Canonical Form.

Positive form

bilinear form

Matrix Analysis, (Matrix Norm)

Determinant.

Fourier series

Integral and differential form

More topics in series and sequence of functions.

Differential equation.

Goal

1. All Problem in:

Drumitt,

Hoffman,

Reubin, KKG

Munkres,

Hutcher,

O'neil,

Silverman.

Solve, and optimize in my own language.

Every study papers should be written under the fixed rules:

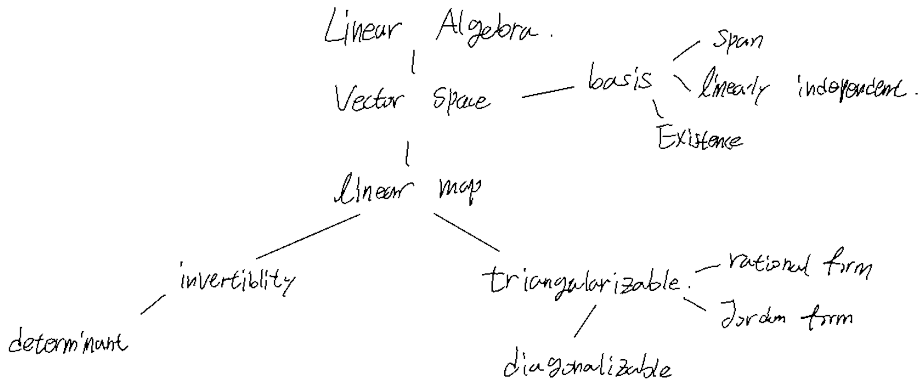
Prop: Proposition. A useful fact

Lem: Lemma.

Thm: Theorem. A main result in the topic

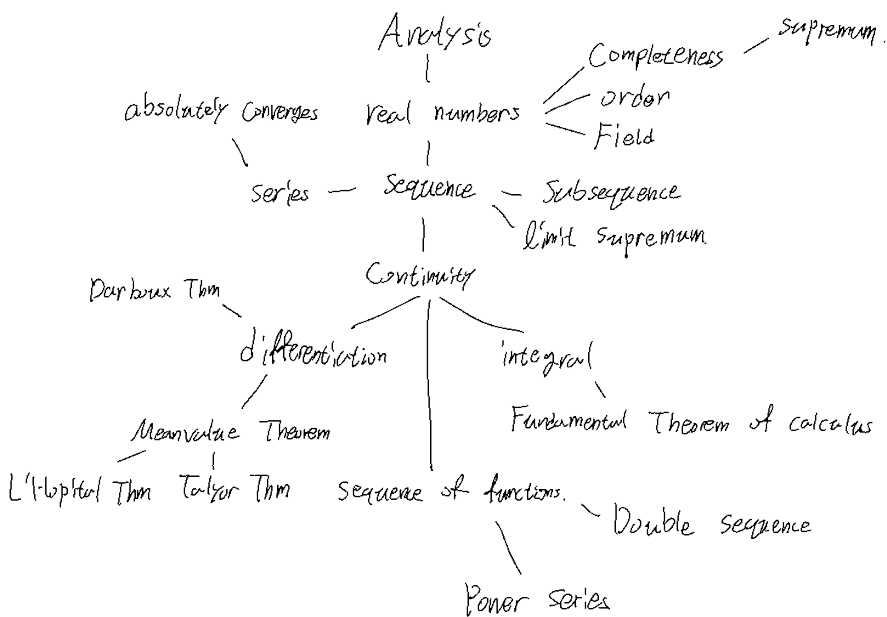
Cor: Corollary.

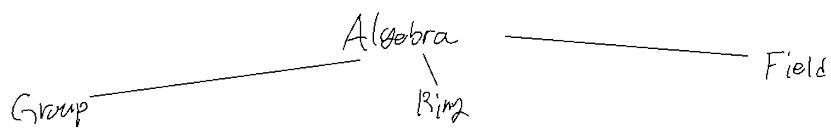
Main Process to understand:



Linear Algebra - View of Geometry

- Inner Product.





Topology

Complex Analysis

Complex number — Completeness
Field

Complex function

differentiation

Cauchy-Riemann equation.

analytic

Series representation

integral

Cauchy integral formula

Algebraic Topology

Differential Geometry

